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SUPERSEDING
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MILITARY SPECIFICATION

MOLDING MACHINES, MEAT-FISH

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers manually and electrically operated, bench mounted and floor model meat and fish molding machines.

1.2 Classification.- Meat and fish molding machines shall be of the following types and classes as specified (see 6.2).

Type I - Manually operated, a minimum of 40 patties per minute
Type II - Electric motor operated, self contained, bench mounted, with automatic paper feed.

Class 1 - Light duty, minimum 1200 patties per hour.
Class 2 - Medium duty, minimum 2100 patties per hour.

Type III - Electrically operated, for attachment to electric motor operated meat grinder; incorporating an automatic paper feed minimum 2400 blocks per hour.

Type IV - Electric motor operated, self contained, floor mounted, minimum 3600 patties per hour, with automatic counter, stacker, automatic paper feed, and automatic conveyor.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-636 - Motor, Alternating-Current, (Fractional Horsepower)
FF-C-77 - Casters; Institutional Duty

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OO-G-667	-	Grinder, Meat, Electric, and Grinder, Meat, Hand
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	-	Preservation, Methods Of
MIL-B-121	-	Barrier Material, Grease-Proofed, Waterproofed, Flexible
MIL-V-173	-	Varnish, Moisture and Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-T-22085	-	Tapes, Pressure-Sensitive Adhesive, Preservation and Sealing

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirement for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurements Of
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

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American Society for Testing and Materials (ASTM)

Specification A 167-70 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W. Washington, D.C. 20036.)

National Sanitation Foundation (NSF)

Standard No. C-2 - Basic Criteria for the Evaluation of Special Equipment and/or Devices
Standard No. 8 - Commercial Powered Food Preparation Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, Michigan 48106.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

Underwriters Laboratories, Inc. (UL)

Standard No. 73 - Motor-Operated Appliances
Electrical Appliances and Utilization Equipment List

(Application for copies should be addressed to the Underwriters Laboratories, Inc., Walt Whitman Road, Melville, L.I., New York 11746; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), the supplier shall furnish a sample molding machine for first article inspection and approval (see 4.2 and 6.3).

3.2 Standard product.- The molding machines delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with specification requirements.

3.3 Standards.- Molding machines except the type I molding machine, shall conform to NSF Standard No. 8. Type I molding machines shall conform to the applicable requirements of NSF Standard No. C-2. The type II, type III, and type IV molding machines shall conform to the applicable requirements of UL Standard No. 73.

3.3.1 NSF certification.- The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the molding machine he proposes to furnish under this specification meets the applicable requirements of NSF Standard No. 8 as follows:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished molding machine shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

(2) A certification for the molding machine issued by NSF under their special one-time contract evaluation/certification service shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the molding machines are being procured, indicating that the molding machines have been tested and comform to NSF Standard No. 8, shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to commencing production.

3.3.2 UL certification.- Prior to approval of the preproduction sample if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the types II, III, and IV molding machine he proposes to supply under this specification meet the applicable requirements of UL Standard No. 73. Acceptable evidence of meeting the applicable requirements of this standard shall be the UL label; UL Listing mark or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the type II, type III, and type IV molding machines have been tested and conform to UL Standard No. 73.

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3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial molding machine provided the completed item complies with all provisions of this specification. Commercial tolerances are applicable to all material thickness.

3.4.1 Corrosion-resisting steel.- The corrosion-resisting steel shall conform to types 302 or 304 of ASTM Specification A 167.

* 3.4.2 Casters.- Casters shall conform to type I (rigid type) and type II (swivel type), class B, wheel diameter 5 inches, style C, mounting optional of FF-C-77.

* 3.5 Design and construction.- Types I and II molding machines shall be designed for counter or bench mounting; the type III molding machine for quick attachment to the larger sizes of meat grinding machine (see 3.5.3); and the type IV molding machine shall be of the floor model type. Molding machines shall be of the dimensions and hopper capacities specified in table I. Type I, II and IV molding machines shall consist of a meat hopper, base, mold plate, mold ejection mechanism, and mechanisms that manually or automatically, as applicable, feed, mold, and eject ground meat or fish as 3 ounce, 4-inch diameter patties. When specified (see 6.2), the type II and type IV molding machine shall be provided with an additional mold plate to produce entree size patties (6 ounce, 1-inch thick), meatballs (approximately 1-1/2 inches diameter) and round patties (2 ounce, 1/4-inch thick).

TABLE I.- Overall dimensions and hopper capacity

Types	Length maximum	Width maximum	Height maximum	Hopper capacity minimum
	Inches	Inches	Inches	Pounds
I	24	22	25	15
II				
Class 1	36	13	20	20
Class 2	40	36	37	15
III	22	10	14	N/A
IV	92	36	67	175

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3.5.1 Type I molding machine.- The type I molding machine shall consist of components specified in 3.5 and shall have a hand-operated lever connected to a mechanism that feeds and molds ground meat or fish.

3.5.1.1 Capacity.- Unless otherwise specified (see 6.2), the type I molding machine furnished shall be equipped with molds to produce 3 ounce, 4 inch diameter patties. The type I molding machine shall be capable of producing patties at a minimum rate of 40 patties per minute when tested as specified in 4.4.2.1.

3.5.2 Type II molding machine.- Type II molding machines shall consist of components specified in 3.5 and shall be supplied in two classes that incorporate an automatic paper feed unit that inserts a paper separator between each patty.

3.5.2.1 Capacity.- Unless otherwise specified (see 6.2), the type II, classes 1 and 2 molding machines shall be equipped to produce 3 ounce, 4 inches diameter patties. The type II, class 1 and 2 molding machine shall have a minimum rate of 1200 and 2100 patties per hour respectively, when tested as specified in 4.4.2.2.

3.5.3 Type III molding machine.- The type III molding machine shall be designed for quick attachment to the larger sizes and styles of the type II meat grinding machine of 00-G-667. The type III molding machine shall consist of a body, molding device, and an integral automatic paper feed unit that inserts a paper separator between each meat or fish block. Type III molding machines shall be supplied without an automatic stacker unit. Molded blocks shall be square or rectangular in shape.

3.5.3.1 Capacity.- The type III molding machine shall be adjustable to produce from a minimum rate of 2400 to 3600 blocks of meat or fish per hour weighing 1/2, 1, or 1-1/2 pounds when tested as specified in 4.4.2.3.

3.5.4 Type IV molding machine.- The type IV molding machine shall consist of the components specified in 3.5, an automatic counter-stacker unit, automatic conveyor or take away assembly as specified in 3.5.4.3 and 3.5.4.4, respectively, and a cabinet.

3.5.4.1 Capacity.- Unless otherwise specified (see 6.2), the type IV molding machine shall be equipped to produce 3-ounce, 4 inch diameter meat or fish patties. The output of the type IV molding machine shall be a minimum rate of 3600 patties per hour when tested as specified in 4.4.2.2.

3.5.4.2 Cabinet.- The cabinet inclosure for the type IV molding machine shall be of the inclosed or open type, fabricated of corrosion-resisting steel specified in 3.4.1 or of corrosion-resistant metal. The cabinet inclosure shall be mounted on four casters, two rigid and two swivel with brakes, or mounted on four legs or mounted on a combination of casters and legs in accordance with the manufacturers commercial practice. The casters shall be as specified in

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3.4.2. The legs, when provided, shall be the adjustable type. When a combination of casters and legs are provided there shall be a minimum of two rigid casters on the same side.

3.5.4.3 Type IV automatic counter-stacker.- The automatic counter-stacker unit shall be a component of type IV molding machine and shall be driven either mechanically by the molding machine mechanism or electric motor. When the automatic counter-stacker unit is supplied with electric motor , a power on-off switch and a power on indicator light shall be provided. The automatic counter-stacker unit shall be a component of the type IV molding machine and shall incorporate an automatic paper feed unit that inserts paper between each patty. The automatic counter-stacker unit shall operate in conjunction with the automatic conveyor or take away assembly (see 3.5.4.4) and shall be capable of counting and stacking from 2 to 20 patties on the automatic conveyor or take away assembly. The automatic counter-stacker unit shall be capable of handling the maximum output of the molding machine specified in 3.5.4.1. Whenever the desired amount of patties are stacked the automatic counter-stacker unit shall cut off and automatically move the conveyor or take-away assembly for the next stack.

3.5.4.4 Automatic conveyor or take-away assembly.- The type IV molding machine shall be provided with an automatic conveyor or take-away assembly. The automatic conveyor or take-away assembly shall be either a separate unit or a component of the type IV molding machine and shall be capable of automatically moving one patty or stacks of counted patties as desired, away from the molding machine. The automatic conveyor or take-away assembly shall consist of a conveyor belt or take-away assembly with mechanisms driven electrically or mechanically. When the automatic conveyor or take-away assembly is supplied with electrical driven mechanism a power on-off switch and a power on indicator light shall be provided. The automatic conveyor or take-away assembly shall work in conjunction with the automatic counter-stacker unit (see 3.5.4.3) and shall be capable of handling the maximum output of the type IV molding machine specified in 3.5.4.1.

3.6 Electrical requirements.-

3.6.1 Electric motor.- Unless otherwise specified (see 6.2), the electric motor for the type II, type III molding machine shall operate on a nominal 120-volt, 60 hertz, 1-phase, alternating-current (A-C). Unless otherwise specified (see 6.2), the electric motor for the type IV molding machine shall operate on a 230-volt, 60-hertz, 1-phase, A-C system. The electric motors shall be totally enclosed or drip proof and shall conform to CC-M-636. The electric motor shall have a minimum of 1/4 horsepower rating for continuous operation under conditions specified in 3.5.2.1 and 3.5.4.1 and shall be thermal overload protected.

3.6.2 Cord and attachment plug.- Types II, III, and type IV molding machine and components i.e., automatic counter-stacker unit and automatic conveyor assembly shall be provided with a 3-prong grounding type attachment plug cap. The cord shall be a 3-conductor grounding type power supply cord, UL type SJ, SJT, or SV. One conductor to be a grounding wire and the cord shall

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be provided with a parallel-blade grounding type attachment plug cap. The grounding conductor of the cord shall be attached to all exposed dead metal parts of the machine at one end and shall be attached at the other end to the ground blade wiring terminal of the attachment plug cap. The cord shall be a minimum of 6 feet in length.

3.6.3 Controls.- An on-off switch shall be provided to energize the circuit for the type II, type III, and type IV molding machine. When the on-off switch is placed in the off position all ungrounded lines shall be broken.

3.6.4 Electromagnetic compatibility.- When specified (see 6.2), types II, III and IV molding machines shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.3).

3.7 Fungus and moisture resistance.- When specified (see 6.2), the machine shall be treated to resist fungus and moisture as follows:

3.7.1 Circuit elements with temperature rise not more than 75 degree Fahrenheit (F).- Circuit elements, which have a temperature rise of not more than 75°F. when operating at full load shall be coated with fungus-resistant varnish conforming to MIL-V-173. Circuit elements include, but are not limited to, cable, wire, terminal blocks capacities, and coils.

3.7.2 Electrical components not requiring fungus moisture resistance.- Electrical components such as switches, fuses and contacts, shall not be treated. Other materials and components which are inherently fungus resistant or are protected by hermetic sealing need not be treated.

3.8 Lubrication.- Means shall be provided except sealed for life bearings, for lubricating bearings, shafts and other moving parts requiring lubrication. At the time of delivery, the machines shall be completely lubricated with the manufacturer's standard lubricant.

3.9 Welding and soldering.- Exposed welds shall be sound and ground smooth. Welding and soldering, as applicable, shall be continuous, smooth and free from burn holes, cracks, flash, fractures, imcomplete fusion, porosity, pin holes or under cuts. All slag or flux deposits (when flux is used) shall be removed from the welds or soldered joints as applicable.

3.10 Repair and maintenance.- Provisions shall be made to allow for ready adjustment, service, and replacement of components without disturbing or disassembly of any major components. In addition there shall be free access to inspect, service and adjust any component or equipment as provided for by this specification or as indicated by the component or equipment manufacturer, without requiring disassembly of any other component or equipemnt on the molding machine.

3.11 Operation.- All molding machines shall be operationally tested as described in 4.4.1 for conformance with the following requirements prior to the test specified in 4.4.2:

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(a) Proper operation of adjusting devices (as applicable).
(b) Proper operation of starting and stopping devices (as applicable).
(c) Proper operation of components i.e. automatic counter-stacker unit, automatic paper feed unit and conveyor assembly (as applicable).
(d) No undue vibration or noise.
(e) No overheating of bearings.
(f) Proper operation of parts.
(g) No binding.

3.12 Functioning.- Each molding machine shall be functional tested as specified in 4.4.2 for the following requirements:

(a) Any nonconformance of uniformity and capacity, production rate, size, and weight of patties or blocks, as applicable and when applicable size, thickness and shape.

(b) Any evidence of discoloration of patties or odors, indicating meat or fish has been subjected to excessive pressure. The number of broken patties or blocks shall not exceed 5 percent.

3.13 Marking for identification.- The molding machines shall be permanently marked with the manufacturer's name or trade name or trademark of such known character as to be readily identifiable with the manufacturer, applied in accordance with MIL-STD-130.

3.14 Finish.- The finish of the molding machine and cabinet shall be in accordance with the requirements of NSF Standard No. 8. Exposed surfaces shall be free from discoloration, rust or stains.

3.15 Workmanship.- The finished molding machines shall be clean and shall not contain any burrs, slivers, sharp edges, rough die, tool, gouge, trim or grind marks. All kinks, bends and winds shall be removed from sheet metal or structural sections of the molding machine cabinets. Castings shall be of uniform quality, and shall not have any cracks, hard spots, blow holes or be porous. Threaded fasteners or rivets, as applicable, shall not be missing, stripped, broken, loose, cracked, wrong size or inadequately peened.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article.- When a preproduction sample is required, it shall be examined for the defects listed in table II, and for dimensions specified, and tested as specified in 4.4.1, 4.4.2 and 4.4.3, as applicable. The presence of any defects listed in table II or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 End product inspection.- The inspection lot shall consist of all molding machines of one type and class offered for inspection at one time. The sample unit shall be a completely fabricated molding machine.

4.3.2.1 Visual examination.- Examination shall be made of the end item for the defects classified in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified.		X
	Edges and surfaces not smooth.	X	
	Evidence of discoloration, rust, pits, scale or stains.	X	
Construction and workmanship (applicable to all components and assemblies)	Component missing.	X	
	Kinks, bends or winds not removed.		X
	Any burr, slivers, sharp edges, rough die, tool, gouge, trim or grind marks.	X	
	Any component fractured, buckled, bent, punctured, or malformed.	X	
	Component not readily accessible for adjusting, servicing and replacement.	X	

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship (applicable to all components and assemblies)	Adjustable assembly not adjustable.	X	
	Assembly not properly adjusted to perform the function intended.	X	
	Functioning component that requires abnormal force to operate.	X	
	Operation omitted or not properly performed.	X	
Casters (when applicable)	Not as specified.		X
	Not provided with two foot brakes.		X
Legs (when applicable)	Not adjustable type.		X
Castings	Castings not of uniform quality.	X	
	Castings not free from cracks, hard spots, blow holes or is porous.	X	
Welding or soldering (as applicable)	Exposed welds not sound and ground smooth.	X	
	Not continuous, smooth, free from burn holes, cracks, flash, fractures, incomplete fusion, porosity, pin holes, under cuts or slag inclusion.	X	
	Flux not removed.		X
Lubrication	Not lubricated where required.	X	
	Means not provided for lubrication.	X	
Threaded fasteners and rivets (as applicable)	Missing, stripped or broken.	X	
	Loose, wrong size, cracked, or inadequately peened.	X	
Identification marking	Missing, incomplete or illegible.		X

4.3.2.2 Dimensional examination.- Examination shall be made of the end item to determine compliance with the dimensional requirements of this specification. Any dimension that is not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

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4.3.2.3 Testing of end item.- Each molding machine shall be tested as specified in 4.4.1. Any nonconformance shall be cause for rejection of the item. The supplier shall perform the tests specified in 4.4.2 using an inspection level of S-2 and an AQL of 2.5 defects, expressed in terms of defects per hundred units. Any nonconformance shall be cause for rejection of the lot.

4.3.3 Examination of preparation for delivery.- An examination shall be made to determine that preservation and packaging, packing, and marking as required by section 5 are complied with. Defects shall be as indicated in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers in the inspection lot. The inspection level shall be S-4 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examination	Defect
Marking, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence or method of application
Materials	Component missing, damaged, or otherwise defective
Workmanship	Blocking and cushioning inadequate, improper, or missing in fiberboard shipping containers

4.3.3.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636. When shipping containers are required to be in accordance with PPP-B-601 or PPP-B-621, examination for defects in closure and strapping shall be in accordance with the applicable container specification and examination for defects in cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186.

4.3.3.2 Examination of palletization.- An examination shall be made to determine that palletization is in compliance with section 5 of this specification. Defects shall be as indicated in table IV. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

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TABLE IV.- Examination of palletization

Examination	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.3.4 Standards compliance.- Proof of compliance with the requirements of NSF Standard No. 8, as specified in 3.3.1, and UL standards, as specified in 3.3.2 shall be available to the Government representative.

4.4 Tests.-

4.4.1 Operational tests.-

* 4.4.1.1 Type I molding machine.- The lever of the type I molding machine complete with molding parts and without a load of meat or fish shall be operated 25 times to determine compliance with the operational requirements in 3.11. Any noncompliance with the requirements specified shall constitute failure of this test.

4.4.1.2 Type II, III and type IV molding machines.- The type II, classes 1 and 2; type III; and type IV molding machine complete with molding parts and components, i.e., automatic paper feed unit, automatic stacker and counter unit and conveyor assembly, as applicable, shall be connected to a power source and operated without a load of meat or fish for a minimum of 15 minutes to determine compliance with the operational requirements specified in 3.11. The type III molding machine shall be attached to a meat grinder for this test. Any noncompliance with the requirements specified shall constitute failure of this test.

4.4.2 Functional test.- The functional test will be performed after the operational test specified in 4.4.1. Any noncompliance with the requirements specified shall constitute failure of this test.

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* 4.4.2.1 Type I molding machine.- The type I molding machine shall be operated to mold medium grade, raw beef that has been ground twice to determine the production rate, capacity, and size requirements of 3.5.1.1. Any noncompliance with the requirements specified shall constitute failure of this test.

* 4.4.2.2 Type II and type IV molding machine.- The type II, classes 1 and 2, and type IV molding machine, complete with molding parts and components, i.e., automatic paper feed, automatic stacker and counter unit and conveyor assembly, as applicable, shall be connected to a power source to determine compliance with the production rate, capacity, and size requirements of 3.5.2.1 and 3.5.4.1, and functional requirements of 3.12. Type II and IV molding machines shall have a hopper of at least 15 pounds and 35 pounds respectively, of medium grade raw beef that has been ground twice shall be provided for this test. Any noncompliance with the requirements specified shall constitute failure of this test.

* 4.4.2.3 Type III molding machines.- The type III molding machines shall be connected to a power source and attached to an electric motor-operated meat grinder with a hopper full of a predetermined weight of medium grade, raw beef that has been ground once. The meat grinder shall be operated to regrind and feed the ground meat into the molding machine and cause the molding machine to perform the molding operation to determine compliance with the functional requirements in 3.12. Inability to meet the functional requirements specified shall constitute failure of this test.

* 4.4.3 Electromagnetic compatibility.- When electromagnetic compatibility is required, the preproduction sample or initial unit of production shall be tested by the supplier in accordance with test methods CE03, and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan and the Electromagnetic Compatibility Test Report required in MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Disassembly.- The extent of disassembly shall be determined by the manufacturer. This determination shall be based on the need for protection of parts susceptible to damage during transit or storage and for the reduction in cubage. Attaching hardware shall be replaced on one of the mating parts or placed in a cloth drawstring bag and the bag attached to one

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of the mating parts. Parts removed shall be adequately identified or tagged. Where disassembly affects the operation of the machines, warning tags or labels indicating the disassembly shall be attached. Instruction for assembly shall be included either as a part of the warning tag, or label, or as a separate sheet or folder attached to the machines. Disassembly shall not be such that special skills or special tools, other than those furnished, or both are required to place the equipment in operation.

5.1.1.2 Preservation.- All exposed uncoated ferrous metal surfaces of the molding machine, detached parts, and accessories shall be cleaned process C-1, thoroughly dried, and then coated with type P-14 preservative conforming to MIL-P-116. Surfaces from which the preservatives might be removed by contact with packaging and packing materials shall be wrapped with greaseproof barrier material conforming to type I or II, grade A of MIL-B-121. The greaseproof wrap shall be secured by taping, tying, or heat sealed when heat sealable material is used.

5.1.1.3 Sealing.- All openings into the motors, control boxes, switches, et cetera shall be completely sealed with tape conforming to type I of MIL-T-22085.

5.1.1.4 Packaging.- All levers and rotating and moveable components not disassembled or removed that may cause damage or be damaged shall be secured in place to prevent movement during shipment. The detached parts of the molding machine and accessories, preserved as specified, shall be packaged method III of MIL-P-116 in a snug-fitting fiberboard box or boxes conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Box closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level C.- The molding machines shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.-

5.2.1.1 Type I and type III molding machines.- Each type I or type III molding machine complete with disassembled components, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s or PPP-B-636. Each machine shall be immobilized by means of die-cut spring or spacer pads to prevent movement or damage while in transit. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

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5. olding machines.- Each type II or type IV moldi ckaged as specified in 5.1, shall be packed in a od shipping container conforming to overseas 01 or class 2, style 2 or 4, type 3 load of PPP-B erboard box or boxes containing detached compo unused portion of the shipping container. Cushi and waterproofing shall be in accordance with container shall be provided with skids and lance with the applicable container specificat

5.

5. olding machines.- Each type I or type III moldi assembled components, preserved and packaged as sp ked in a snug-fitting fiberboard shipping conta , type CF, variety SW, or type SF, class domes. Each machine shall be immobilized by means of di to prevent movement or damage while in transit. Each closed in accordance with method II of the appen

5. 6.2), the shipping container shall be V3c, V3s, ed in accordance with PPP-B-636.

5. olding machines.- Each type II or type IV moldi ckaged as specified in 5.1, shall be packed as sp that the containers shall be domestic type and water ed.

5. ines, preserved and packaged as specified in 5.1, to insure carrier acceptance and safe deliv west transportation rate for such supplies. Conta ly with Uniform Freight Classification Rules or N fication Rules.

5. otherwise specified (see 6.2), type I or III moldi fied shall be palletized in accordance with load type pared load shall be bonded with primary and secon ith the appendix of MIL-STD-147. Interlocking of l versing the pattern of each course. If the conta s not conform to any of the pallet patterns specil let pattern used shall first be approved by the cont

5. any special marking required by the contract or o shipping containers shall be marked in accordance

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6. NOTES

6.1 Intended machines covered by this specification are int ng and serving establishments to mold ground meat ized patties or 1/2, 1 or 1-1/2 pound blocks.

6.2 Ordering s should specify the following:

(a) Titl specification.
(b) Type e to be furnished (se
(c) When (see 3.1, 4.2 and 6.3).
(d) When required (see 3.5).
(e) Size ht of patties to be molded, if oth .1, 3.5.2.1, and 3.5.4.1).
(f) Elec other than specified (see 3.6.1).
(g) When ty is required (see 3.6.4).
(h) When nt of electrical component is req
(i) Sele f preservation and packaging and
(j) When erboard shipping containers are req see 5.2.2.1.1).
(k) When red (see 5.3).

6.2.1 Contra requirements for equipment manuals for the items c should be included in DD Form 1423 Contract Data R in the contract.

6.3 First ar e is required, it shall be inspected and approved un ns of ASPR 7-104.55. The first article should The first article should consist of one complete acting officer should include specific instru structments regarding arrangements for inspection ticle.

6.4 Supersed 1 and style 2, and type III, class 1, model A and B m from this specification.

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I of MIL-T-

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be secured in
the molding
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B-636. Box

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The supplier

see 6.2).

type III
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6.5 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - MD
Navy - MC, SA

Preparing activity:

Army - GL

Project No. 7320-0365

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MOLDING MACHINES, MEAT-FISH MIL-M-40163C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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